

Detecting Operations Research patterns in Automated Planning domains

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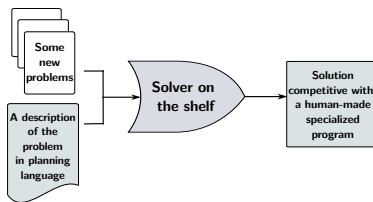
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1 Introduction

- Generates a sequence of actions to reach a **goal** given an **initial state**.



Limitations

- Scaling up complex domains causes search spaces to grow **exponentially**.
- Classical planners can **time out** on large problems.

Objectives

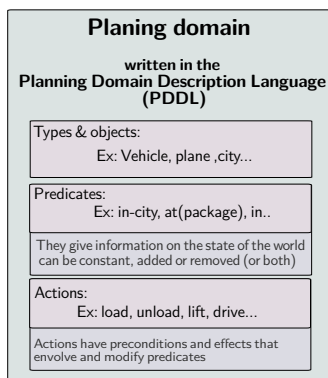
- Identifying possible hidden **Operations Research (OR)** patterns within standard domains.
- Exploiting this knowledge to *potentially* improve performance

Contributions

- A Matrix categorizing hidden OR problems within **International Planning Competition (IPC)** benchmarks
- A **"Divide and Conquer"** algorithm that leverages this structural knowledge to solve problems.

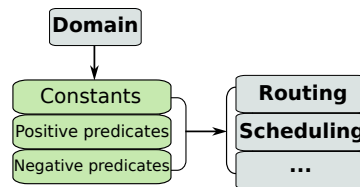
2 Approach & Methodology

Predicates structure



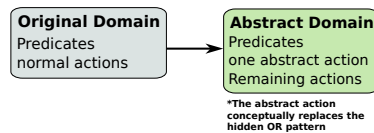
OR Pattern detection Predicates analysis

- Constants** identify the static topology of the problem.
- Positive predicates** track entity progression
- Negative predicates** can model constraints and limits

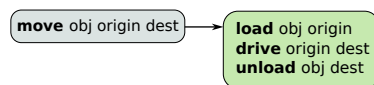


3 Script Implementation

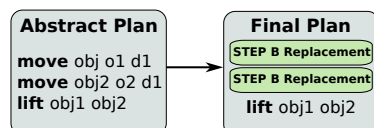
- Step 1: Global resolution of the problem with an Abstract domain using Fast Downward



- Step 2: For each abstract action of the plan, generate a sub problem and resolve it



- Step 3: Plan reconstruction -> replacing all the abstract actions with their respective plan



References

- Helmert, M. (2006). *The Fast Downward Planning System*.
- Teghem, J. (2012). *Recherche opérationnelle. Tome 1 : méthodes d'optimisation*. Ellipses.

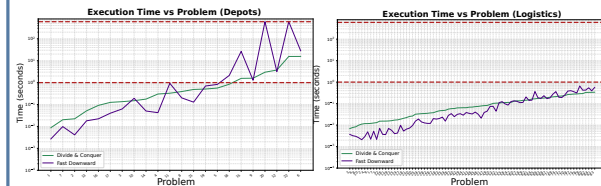
4 Contributions

IPC domains comparison table, categorizing problems by the OR Family.

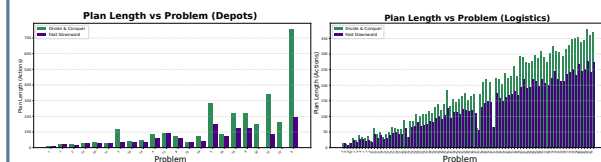
	Graph Problem	Routing	Scheduling	Multi-Agent Path Finding
Blocks				
Elevator		PDP		
Freecell				
Logistics	Minimal feedback	PDP		
Schedule	Sort+Colorability		Open Shop	
Depots		PDP		
Driverlog		VRP/PDP & synchro		
Rovers		HC-VRP		
Satellite	Colorability		unrelated machines	
Zenotravel		PDP		
Airport				Spatial & kinetic Constraints
Pipes world			Pipeline Scheduling	
Openstacks			RCPSP	
TPP		TPP		
Trucks		VRP with time And spatial		

Experimental Results

Presenting results from two of the domains that were worked on, Depots and Logistics, first comparison by speed and then by plan length



- For the Depot domain, our Divide and conquer approach found solutions for all 22 problems, while FD found for only 20 (and timed out for the others)
- Our approach generally performs better when faced with more complex problems where FD times out
- For very simple problems, FD is faster since our Script performs extra steps



- The current major limitation is that the plan length is almost always longer

Future Work

- Completely Automating the process (with automatic OR Pattern detection) using machine learning techniques
- Expanding testing on all IPC benchmark domains